

KONSEP PENGOLAHAN CITRA

MATA KULIAH PENDUKUNG

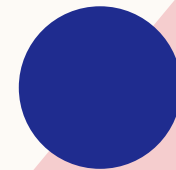
Algoritma Pemrograman

Struktur Data

Matrik & Aljabar Linier

Kalkulus

Komputer Grafik



MATERI :

- Pengertian Pengolahan Citra
- Kuantitas dan Kualitas Citra
- Jenis-jenis citra dan Konversi citra
- Operasi ketetanggaan piksel dan histogram citra
- Operasi Filter Median, Mean dan Rata-Rata
- Operasi Geometri Citra
- Operasi Morfologi
- Segmentasi Citra
- Ekstraksi Citra

REFERENSI :

1. Acharya, T.; Ray, A.K. 2005; *Image Processing Principles and Applications*; New Jersey: John Wiley & Sons, Inc.
2. Agoston, M.K.; 2005; *Computer Graphics and Geometric Modeling Implementation and Algorithms*. London; Springer-Verlag London Limited.
3. Al Tameemi, A.M. Zheng, L. and Khalifa, M. 2011. *Off-Line Arabic Words Classification using Multi-Set Features*. on *Information Technology Journal*, Vol. 10, pp. 1754-1760.
4. Aribowo, A.S. 2009. *Model Penelusuran Citra Digital pada Database Citra Menggunakan Pendekatan perhitungan Kedekatan Pola Warna*. Prosiding Seminar Nasional Teknik Informatika e-Democracy, UPN Yogyakarta.
5. Boncelet, C. 2009. *Image Noise Models*; on *The Essential Guide to Image Processing*. A.L. Bovik (Editor). London: Elsevier Inc.
6. Bourne, R.; 2010. *Fundamentals of Digital Imaging in Medicine*. Springer-Verlag London Limited.
7. Bradley. D.; and Roth, G. 2007. *Adaptive Thresholding using the Integral*. On *Journal of Graphics, GPU, and Game Tools*. Vol.12, No. 2, pp. 13-21.
8. Burger, W, and Burge, M.J. 2008 *Digital Image Processing An Algorithmic Introduction using Java*. New York: Springer Science+Business Media, LLC.

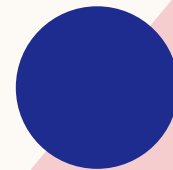
PENILAIAN :

- UTS : PRESENTASI MATERI
- UAS : MEMBUAT PROGRAM APLIKASI
- DLL

CITRA = GAMBAR

Representasi Obyek Dunia

Nyata pada media 2 dimensi







COMPUTER
VISION

PENGENALAN POLA

PENGOLAHAN CITRA DIGITAL

INPUT CITRA



(a) Kamera digital



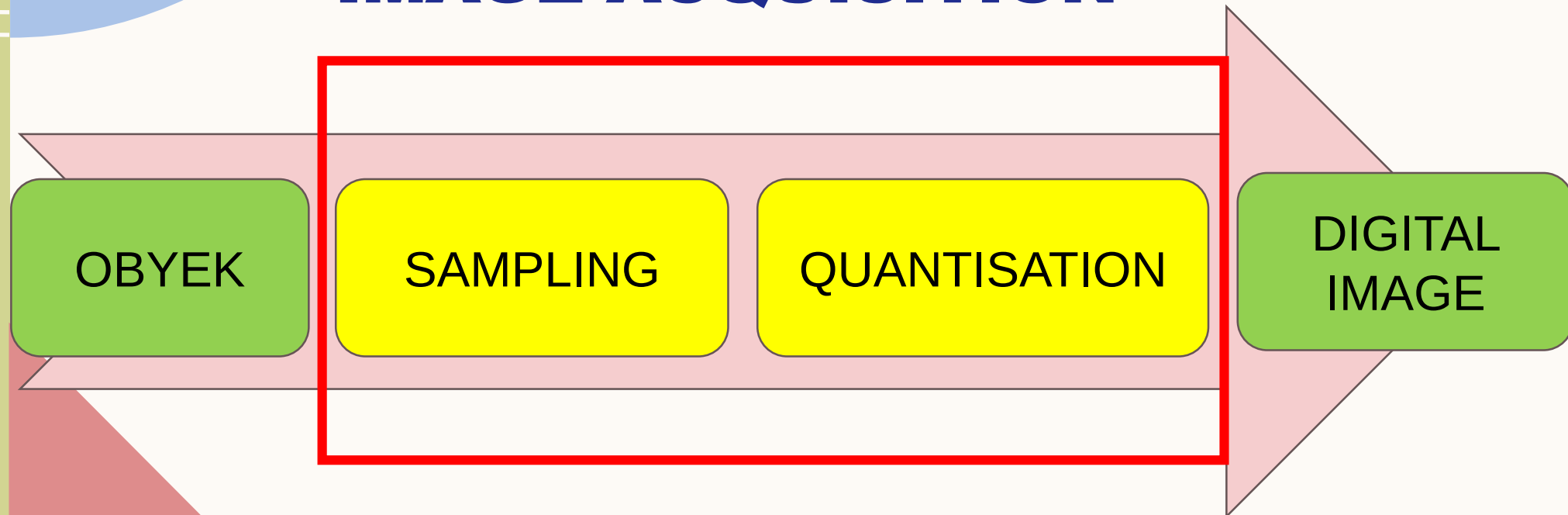
(b) Kamera CCTV



(c) Pemindai (Scanner)

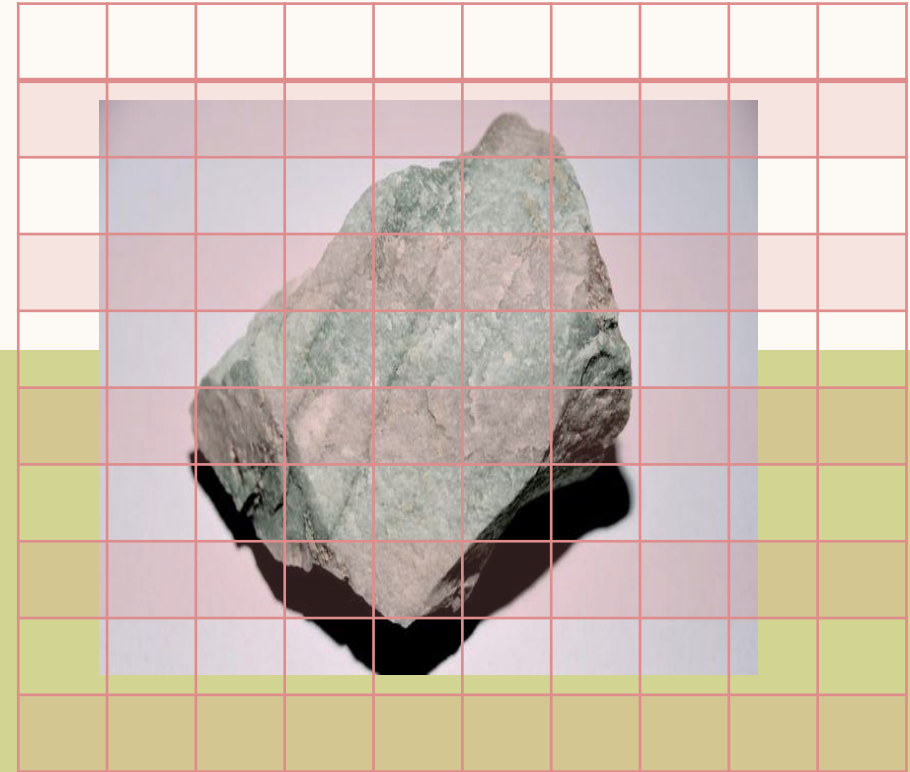
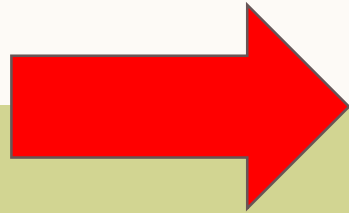


IMAGE ACQUISITION

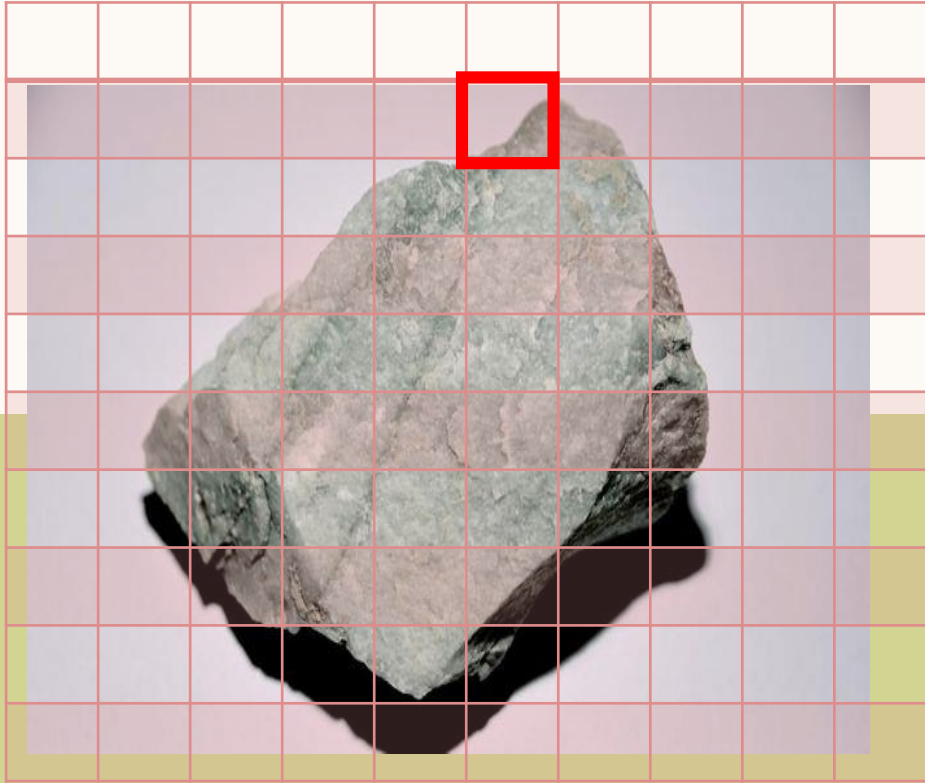




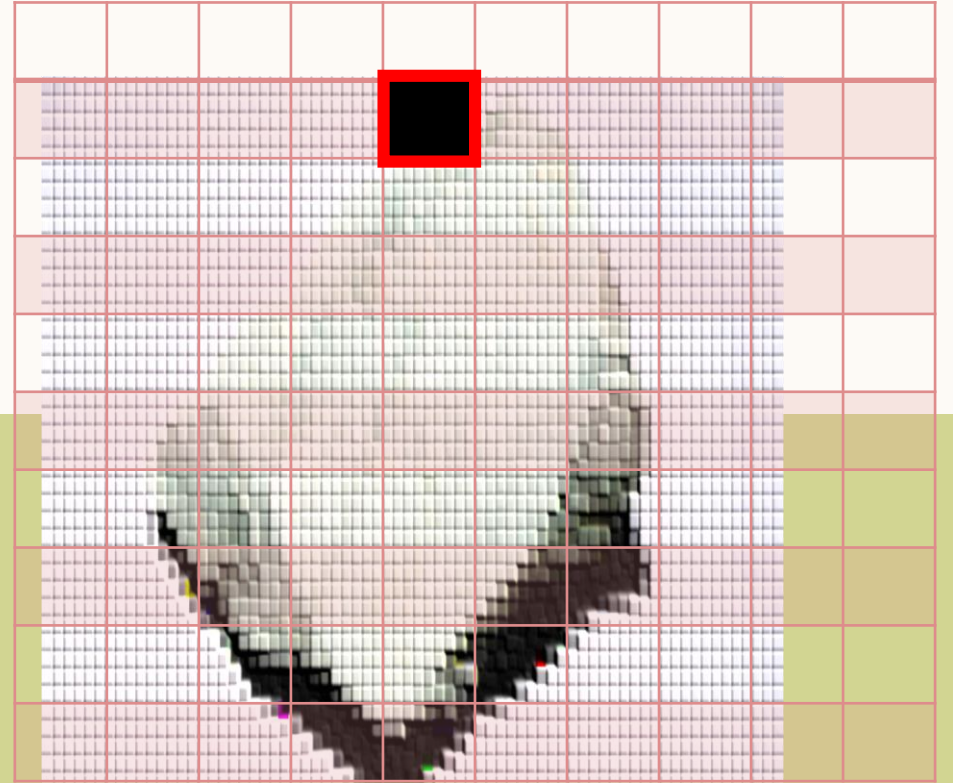
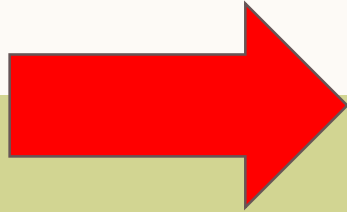
Obyek Dunia Nyata



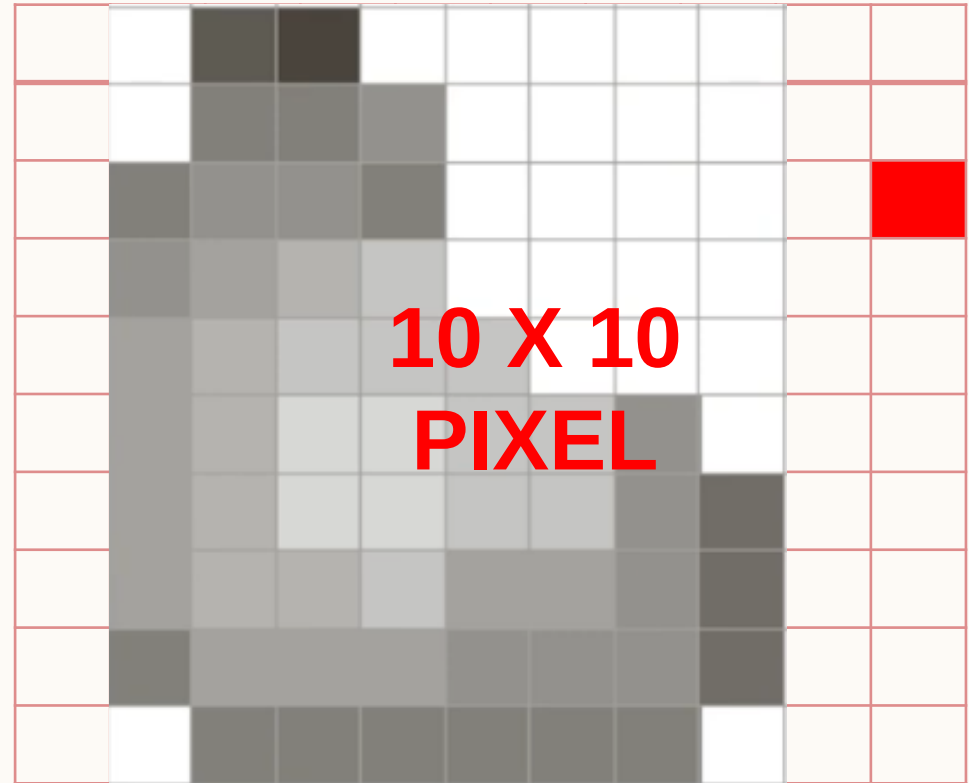
Proses Sampling



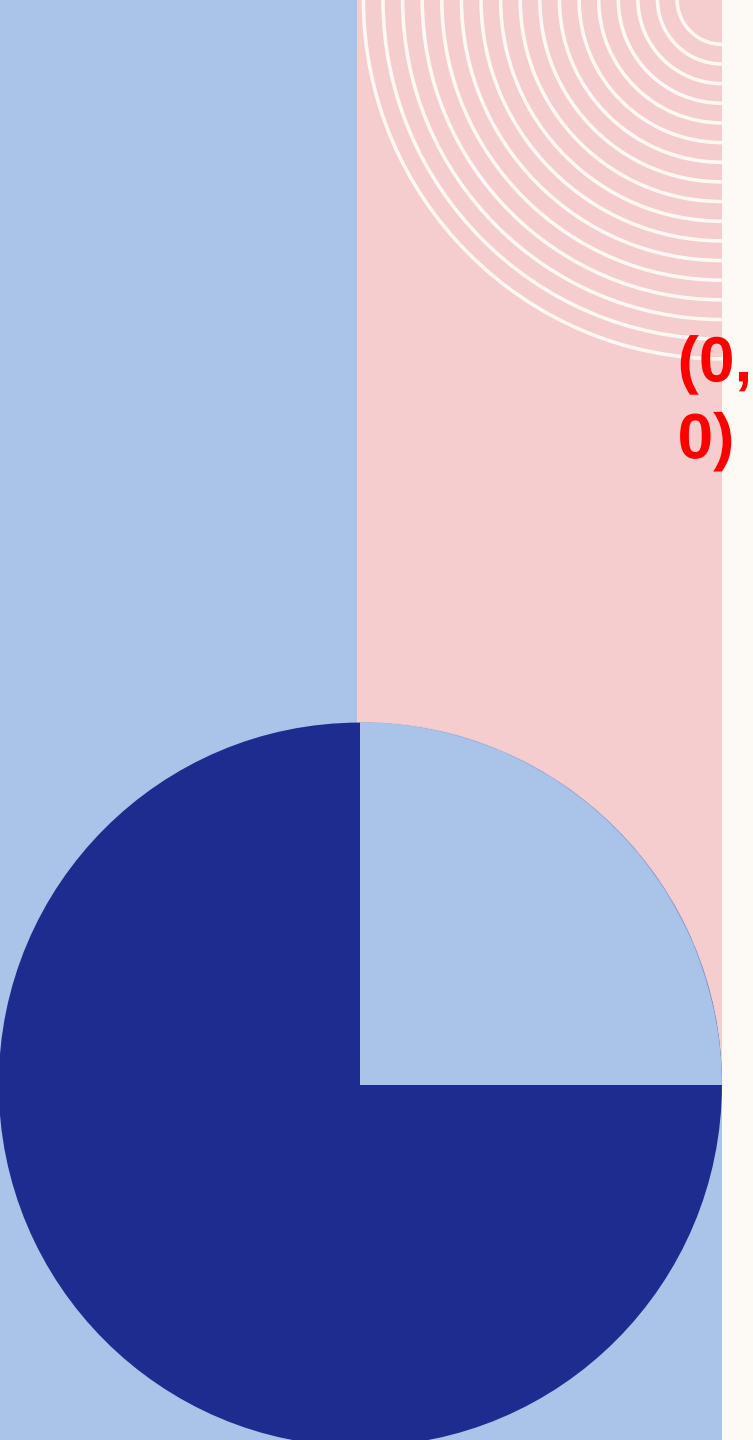
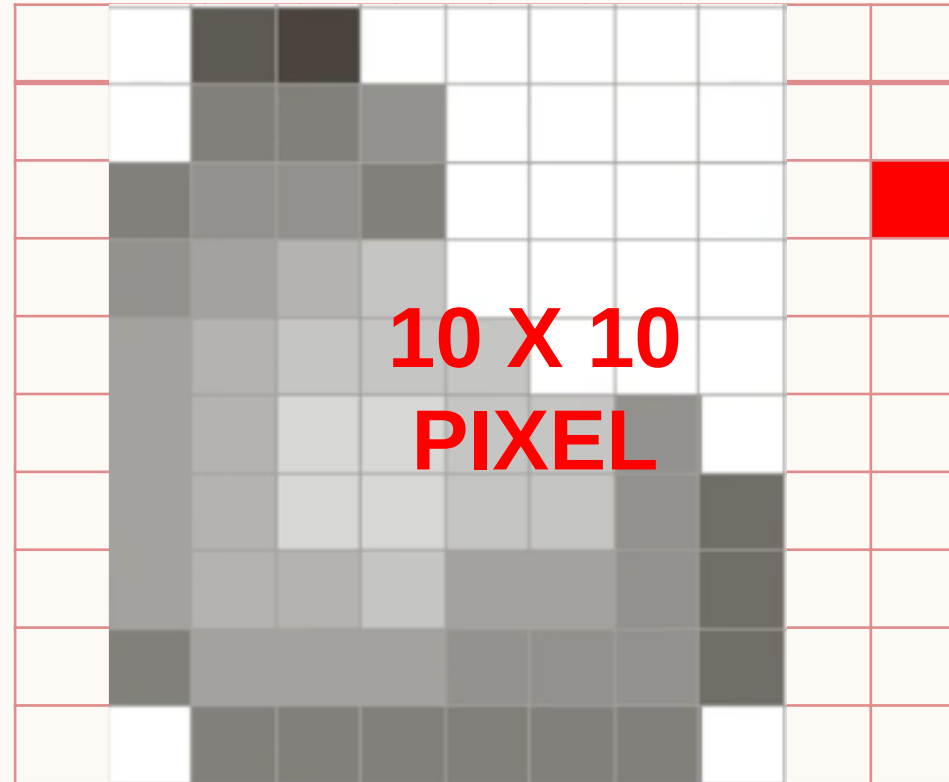
Proses Sampling



Citra Digital



PIXEL

 $(0, 0)$ 

10 X 10
PIXEL

PIXEL

(PIXEL ELEMEN)

 (9.9)

256 x 256 pixel



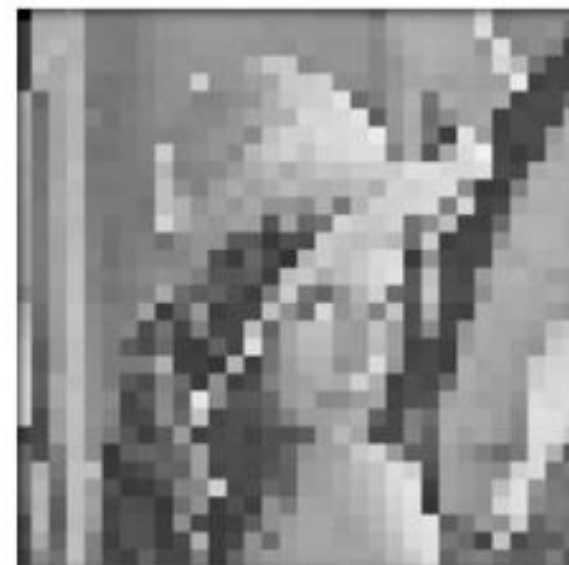
128 x 128 pixel



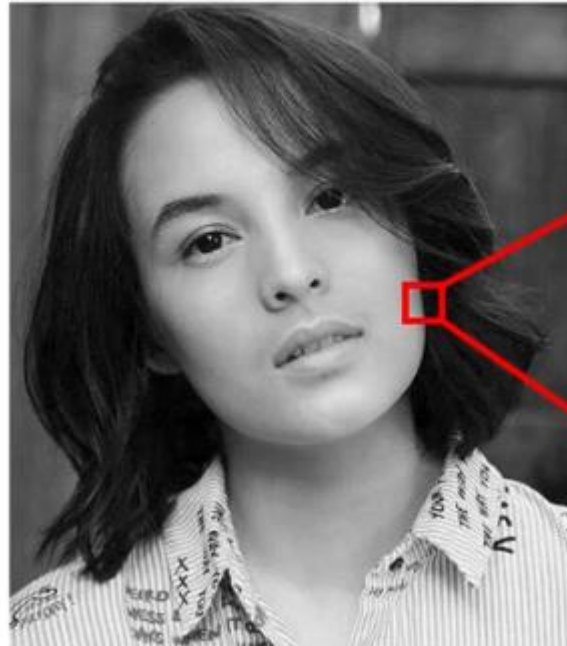
64 x 64 pixel



32 x 32 pixel



Yang dilihat manusia



Yang dilihat komputer

200	50	0	0	0
226	80	7	0	0
255	237	201	0	0
255	230	200	0	0
255	220	108	5	0

MATRIK

THANK YOU